Machine Learning

Applied Computational Intelligence (sem. 1)

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I. Aims of the activity

- To approach the field of Machine Learning from a broad, breadth-oriented perspective, primarily a research-oriented one.
- To cover the principles, good practices, design, and evaluation of learning systems that improve their performance on a set of tasks through experience.
- To offer a broad understanding of fundamental machine learning algorithms and their use in data-driven knowledge discovery.
- To get familiarized with the methodology of research in the field of Machine Learning, by:
    a. Exploring various learning paradigms, the specific characteristics of each paradigm, and potential research directions within the field.
    b. Broadening the perspective on ML and developing a comprehensive understanding of its different approaches, challenges, and research opportunities
    c. Offering an understanding of the current state of the art in machine learning to conduct original research in the field.

II. Specific competencies acquired

Professional competencies

1. Understanding the concepts, methods, and models used in Machine Learning.
2. Understanding the principles, design, implementation, and validation of learning systems.
3. Learning to conduct original research in machine learning.

Transversal competencies

1. The ability to apply machine learning techniques in solving real-world problems.
2. Responsible execution of assignments, research and practical reports.
3. Application of efficient and rigorous working rules.
4. Manifest responsible attitudes toward the scientific and didactic fields.
5. Respecting the professional and ethical principles.

III. Course content

1. Introduction to Machine Learning. Statistical Foundations.
2. Supervised learning.
   2.1. Tree-based learning.
   2.2. Neural network-based learning.
   2.3. Support Vector Machines.
   2.4. Bayesian learning.
   2.5. Instance-based learning.
3. Unsupervised learning.
4. Reinforcement learning. Other learning models and paradigms.

IV. ML activities

All activities require physical participation.

- The course materials are available in **General/Shared/Class Materials**.
- The lecture hours in Weeks 9-14 will be allocated for research report presentations (face-to-face), according to the time planning available in **TimePlanningReport.docx.**
- The lab assignments must be submitted on MSTeams (through the corresponding Assignments) before the lab class and presented during the lab class.
- The research reports must be submitted on MSTeams (through the corresponding Assignments) at least 48h in advance of the presentation date.

Time planning

Lectures

Weeks 1-8 – lectures on the topics from Section III
Weeks 9-14 – research report presentation

Labs

Lab 2 (Weeks 3/4) - Establishing the 2-person teams for the software project
Labs 3 – 6 (Weeks 5/12) Lab assignments presentation (face-to-face)
Lab 7 (Weeks 13/14) - Software projects presentation (face-to-face)

V. Bibliography

1. Mitchell, T., *Machine Learning*, McGraw Hill, 1997
   (available at www.cs.ubbcluj.ro/~gabis/ml/ml-books)
2. Nillson, N., *Introduction to Machine Learning*, Stanford University, 1996
   (available at www.cs.ubbcluj.ro/~gabis/ml/ml-books)
3. Sutton, R.S., Barto, A.G., *Reinforcement learning*, The MIT Press Cambridge, Massachusetts, London, England, 1998 (http://incompleteideas.net/book/the-book.html)
4. Ian Goodfellow, Yoshua Bengio, Aaron Courville, *Deep Learning*, MIT Press, 2016 (online edition at http://www.deeplearningbook.org/)
5. Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, *Dive into Deep Learning*, 2020 (http://d2l.ai/)

6. Li Deng and Dong Yu, *Deep Learning*. *Methods and Applications*, Foundations and Trends® in Signal Processing, Volume 7 Issues 3-4, 2014 (https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/DeepLearning-NowPublishing-Vol7-SIG-039.pdf)
7. Cristiani, N., *Support Vector and Kernel Machines*, BIOwulf Technologies, 2001

Journals

1. *Journal of Machine Learning Research* is a freely available WoS journal - http://jmlr.csail.mit.edu/
2. *Machine Learning* - http://www.springer.com/computer/artificial/journal/10994
3. *Expert Systems with Applications* - https://www.journals.elsevier.com/expert-systems-with-applications
4. *Neural Computation* - http://www.mitpressjournals.org/loi/neco
5. *Neural networks* - http://www.elsevier.com/wps/find/journaldescription.cws_home/841/description#description
6. *IEEE Transactions on Pattern Analysis and Machine Intelligence* - *http://www2.computer.org/portal/web/tpami/*
7. *Bioinformatics* is a journal focusing on analysing biological data - http://bioinformatics.oxfordjournals.org/
8. others

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Conferences

International Conference on Machine Learning, Neural Information Processing Systems, Conference on Computational Learning, International Conference on Knowledge Discovery and Data Mining, European Conference on Machine Learning, and others.

Resources

- ML repositories
  - http://archive.ics.uci.edu/ml/
  - https://www.kaggle.com/datasets
- RL repository
  - http://www-all.cs.umass.edu/rlr/
- RL resources
  - https://aikorea.org/awesome-rl/
  - http://busoniu.net/repository.php
- SBSE repositories
  - http://promise.site.uottawa.ca/SERepository/
  - http://openscience.us/repo/
- Object recognition data sets
  - https://cvgl.stanford.edu/teaching/cs231a_winter1314/lectures/datasets.pdf
- Computer vision image datasets
  - http://www.cs.utexas.edu/~grauman/courses/spring2008/datasets.htm
- Other data sets
  - https://vincentarelbundock.github.io/Rdatasets/datasets.html

VI. Activity

Two components are required: a **research report** (**Section VII**) and a **software project** (**Section VIII**).

VII. Research report

Each student will **individually** prepare a **research report** on an ML-based topic/technique, based on recent research papers.

a) a written paper of 9-10 pages using the LaTeX template available in the folder General/Shared/Class Materials/LaTeX template (research paper)

- The use of the LaTeX template is compulsory.
- Overleaf - LaTeX editor (online)
- TeX Live - https://www.tug.org/texlive/ (offline)

b) an oral presentation

- ppt/tex (max **6-7 min**) + Q&A (**5 min**)
- a one-page **outline** of the presentation

Requirements

The research report will present a survey of some recent research papers (more than 6 titles) on the chosen topic.

The report should fulfill the requirements of a research paper:

- suggestive title corresponding to the content;
- about 15-20 lines abstract;
- introductory section, introducing the topic and detailing the purpose and structure of the paper;
- a section presenting the importance and relevance of the chosen topic from a theoretical and practical perspective (application domains);
- main sections presenting recent approaches and existing experimental results on the chosen topic;
- a discussion section which analyses the literature approaches reviewed in the previous sections from a critical perspective (evaluation of methodology, data sources, results, and findings);
- concluding remarks and further work section;
- bibliography of more than 6 titles; the bibliography entries should be written **correctly and completely**; all the bibliography items must be cited in the text.

E.g.,

- Manevitz, L., Yousef, M., 2007. One-class document classification via neural networks. Neurocomputing 70, 1466–1481 (**@article** bibtex entry)
- Le, Q., Mikolov, T., 2014. Distributed representations of sentences and documents, in: Proceedings of the 31st International Conference on Machine Learning, Beijing, China, 2014, pp. 1188–1196 (**@inproceedings** bibtex entry)

- Goodfellow, I., Bengio, Y., Courville, A., 2016. Deep Learning. MIT Press (**@book** bibtex entry)
- Google. Google Code Jam Competition. https://codingcompetitions.withgoogle.com/codejam. Online; accessed 15 September 2021 (**@misc** bibtex entry)

!!!! Do not use references to arXiv articles, as they are not peer-reviewed.

Written paper

- The written paper will have to be submitted at least **48h** in advance of the presentation date.
  - The research reports will be checked against plagiarism.
  - The maximum allowed similarity score is 25%.
  - The similarity report generated using Turnitin software should be submitted, together with the research report in a .pdf format, on MSTeams (through the corresponding Assignments).

Use of generative AI and AI-assisted technologies

Elsevier's AI author policy

- "Authors are allowed to use generative AI and AI-assisted technologies in the writing process, but **only to improve the language and readability of their paper** and with the appropriate **disclosure."**
- **"Authors must disclose the use of generative AI and AI-assisted technologies in the writing process by adding** a statement at the end of their manuscript in the core manuscript file, before the References list. The statement should be placed in a new section entitled 'Declaration of Generative AI and AI-assisted technologies in the writing process'.

  Statement: During the preparation of this work the author(s) used [NAME TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

!!! Be aware of the ethical aspects of using GenAI in scientific research!

Elsevier's Responsible AI principles

- 5 principles
  - fairness, transparency and explainability, accountability, privacy

Oral presentation

- The **oral presentation** of the research report is **not compulsory**.
- The scheduling for the research report presentations can be consulted in the file **TimePlanningReport.docx**.

    - the one-page outline of the presentation must correspond both to the written text and to the oral presentation, and should be self-explanatory;
    - the one-page outline is graded only if the oral presentation will be delivered.
- At most **6-7 minutes** will be allocated for the oral presentation + about **5 minutes** for questions and discussions.
- The presentation (PPT, LaTeX) should cover the relevant issues related to the addressed topic.

The scheduling for the oral presentations should be respected. If a presentation is not delivered on the due date, it cannot be rescheduled. In this case, the grade for the theoretical report will include only the grading of the written article.

Deadline

Week 4 – a one-page **presentation document** (submitted through a MSTeams assignment) containing:

- a brief description of the approached topic and its importance (relevance) (4-5 paragraphs);
- the title of the research report;
- bibliographical references (at least 6);
    - the bibliography entries should be written **correctly and completely**;
    - all the bibliography items must be cited in the text.

Notes

1. For selecting a presentation slot, please include your name and report title in the document **General/Shared/TimePlanningReport.docx**, by the end of **Week 4**.
2. The number of presentation slots available and marked in the document **cannot** be changed.

Grading

- Identical reports are **NOT** allowed.
- The **score** for a **research report** is composed as follows:
    - **5%** the grade for the **presentation document**;
    - **95%** the grade for the written paper and the oral presentation (if the case) computed as follows:

        - 0.5 p = one-page outline (graded only if the oral presentation will be delivered);
        - 0.75 p = abstract, introduction, and title (corresponding to the content);
        - 2.5 p = main sections (including experimental results);
        - 1 p = discussion section;
        - 0.5 p = conclusions and future work;
        - 0.75 p = bibliography accuracy;
        - 1.5 p = quality of oral presentation
        - 2.5 p = question answering

- The delay of **one week** in providing the **presentation document** will be penalized with **1 point**.
- The failure to deliver the presentation on the due date will lead to a grade penalty of **1 point** for each week it is delayed.

VIII. Software project

The **software project** (SP) will have to demonstrate the use of **one** or **two conventional** ML techniques on one (or two) specific learning task(s) and will be developed in **teams of 2 students** (**the teams will be formed at your choice**).

- The **maximum** grade for two-person projects that address **two** learning tasks is **10.**
- The **maximum** grade for two-person projects that address only **one** learning task is **6.**

Requirements

- A **labeled tabular data set** (from existing data repositories) with instances characterized by predefined features will be chosen.
- **One** or **two** learning tasks (**at your choice**) will be **defined** on the chosen data, such as:
    - supervised classification - supervised regression
    - supervised classification/regression - unsupervised classification
    - supervised classification/regression - reinforcement learning
    - semi-supervised classification/regression – supervised classification/regression)
    - etc
- For **each** learning task, a **conventional** ML technique will be chosen, and **two** implementations are required:
    - an implementation **from scratch**, without using existing ML environments
        - the **ML model** (learning algorithm, hyperparameter optimization, etc.) and the **performance evaluation** (cross-validation, evaluation metrics, etc) will be **fully implemented**
        - existing libraries are allowed only for **data visualisation** and **numerical computations** (e.g., numpy)
    - an implementation using an existing **ML software**
        - Python libraries (Scikit-learn, Keras, etc)
        - WEKA http://www.cs.waikato.ac.nz/ml/weka/
        - RapidMiner http://rapid-i.com
        - Orange http://www.ailab.si/orange/
        - ROCKIT http://xray.bsd.uchicago.edu/krl/KRL_ROC/software_index.htm
        - SVM software http://www.support-vector-machines.org/SVM_soft.html
        - MATLAB
        - others

- In the case of **two** supervised learning tasks, the ML methods should be chosen from **different classes** of learning methods (e.g., network-based, tree-based, instance learning-based, etc).

Note

- **statistical learning methods** (e.g., logistic/linear/polynomial/ridge/multiple regression can be chosen, but their implementation (**component (6)**) will be graded with at most **7**.

Software Project components

All documents required for the software project should be provided as .pdf files generated in LaTeX using the template available in the folder General/Shared/Class Materials/LaTeX template (lab assignment)

- **the use of the LaTeX template is compulsory;**
- **for each of the specific requirements (from each component), you should include sections/subsections in your LaTeX document to increase readability.**

The following **components** are required for the software project:

(1) Definition of the **learning task(s)** (doc)
  - problem(s) definition (the statement in natural language, **what** should be solved);
  - problem(s) specification
    - input data
    - description of the input data
    - preconditions
    - output data
    - postconditions;
  - specification of the learning task(s) (**Task**, **Performance**, **Experience**);

(2) Data analysis (doc)
  - analysis of the features used in learning
    - correlation;
    - independence;
    - feature importance;
  - data statistics
  - data distribution
  - data visualization and interpretation

(3) Used **ML technique(s)** (doc)
  - brief description of the employed ML technique(s)
  - design of each of the learning tasks
    - target function to be learned (formal definition)
    - learning hypothesis (approximation of the target function)
    - representation of the learned function
    - learning algorithm

(4) Related work summary (doc)

- a summary of the literature approaches addressing the same learning task(s) and/or employing the same data set, including their performances (obtained results)

(5) Experimental results and discussion (doc);

- details about the ML model(s) implemented from scratch
  - architectures, specific details
  - hyperparameters setting (hyperparameters optimization should be used)
- experimental results - summary (in a tabular form) of the performances of the ML model(s) implemented from scratch
  - for performance evaluation of all ML models
    - cross-validation must be used;
    - a statistical analysis of the results must be provided (e.g., standard deviation, confidence intervals).
  - evaluation measures
    - **for supervised classification**: accuracy, precision, recall, sensitivity, f-measure, AUC (Area under the ROC curve), AUPRC;
    - **for regression**: MAE (Mean of Absolute Errors), RMSE (Root Mean Squared Error), NRMSE (Normalized Root Mean Squared Error), R2
    - **for unsupervised classification (clustering):** both **internal** and **external** evaluation measures
- discussion
  - for each of the ML model(s) implemented from scratch
    - analysis/interpretation of the obtained performance
    - explainability of the model ouputs (using a selected algorithm – LIME, SHAP, etc)
    - comparison to related work
      - summarizing (in a tabular form) the values for the performance metrics for:
        - the model(s) implemented from scratch
        - the model(s) implemented using a library
        - the results from literature (related work)
      - details about the implementations from existing libraries should be given (the models should be tested using the same testing methodology as for the from-scratch implementations)
  - comparative analysis and interpretation of the performances obtained for the two selected learning tasks (if the case)

(6) the electronic version of the source code (.zip archive submitted through the corresponding assignment)

(7) QA session regarding the source code and the project implementation (face-to-face during Lab7) - ONLY if all the required documentations (components 1-6) were delivered.

Deadlines

Lab 2 - Establishing the two student teams (the teams will be included in the documents below):
- **ICA1 group** - General/Shared/Lab/ICA1 Lab - Teams software project.xls
- **ICA2 group** - General/Shared/Lab/ICA2 Lab - Teams software project.xls

Lab 3 – component (1)
Lab 4 - component (2)
Lab 5 – component (3)
Lab 6 – component (4)
Lab 7 – components (5) + (6) +(7)

Grading of the SP

- Identical projects are **NOT** allowed.
- The grade for a component is composed of: **8 points** for the written documentation (submitted through the MSTeams assignment) and **2 points** for a presentation of the component during the lab hours
  - each team member will receive a separate grade (for the written documentation and for the presentation).
- The grade for the **software project** is computed as a weighted **average** of the grades received for the required components (5% component 1; 10% components 2 and 4;15% component 3; 25% component 5, and 35% components 6+7).
- The delay of **one week** in completing a lab assignment will be penalized with **1 point**.

IX. Final grade

The **final grade** is computed as follows:

15% Class attendance and activity
45% Research report
40% Software project

The activities (components of the final grade) **are not compulsory**.

If **components (5) + (6) + (7)** for the software project are not delivered at **Lab7**, they may be delivered in the **examination session**, with a penalty of **2 points** for the delay**.**

- A minimal final grade of 5 is required to pass the course.

X. Semester and final grades

You can view the grades received during the semester, as well as the final grades for the Machine Learning course, at the following link.

The code used for displaying grades is the **unique identification code** from Academic Info.

XI. Retake session

- The activities (research report/software project) can be graded in the retake session, except for the oral presentations, but **ONLY** if at least one activity (research report or software project) has been (partially) completed during the semester.
  - The maximum grade for the **research report** in the retake session is **5.5**.
  - The maximum grade for a **software project** in the retake session is **6**.
- **No** report or software project can be resubmitted for a grade increase in the retake session.

XII. Possible topics for the Theoretical Report (not an exhaustive list)

The topics below are suggestions only. Please feel free to choose other ML/DL related topics.

1. One-class classification
2. Recurrent neural networks
3. Time-delay neural networks
4. Self-organizing maps
5. Hebbian learning
6. Semi-supervised learning
7. Radial Basis Function networks
8. Decision Trees (fuzzy, lazy, etc)
9. Bayesian learning
10. Machine learning in bioinformatics
11. Machine learning in software engineering (*search-based software engineering*)
12. Instance-based learning
13. Case-based learning
14. Boosting algorithms
15. Bagging algorithms
16. (Deep) Q-learning
17. Adaptive clustering
18. (Deep) Support vector machines
19. Kernel methods in machine learning
20. Association rule mining
21. Hidden Markov Models
22. Generative earning
23. Few-shot learning
24. Deep Reinforcement Learning
25. Autoencoders
26. Belief network learning
27. Generative models
28. Siamese networks

29. Contrastive learning
30. Representation learning
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